Exercise 2 **Roll No. 240701385**

Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction using Python (Tkinter for GUI, Speech Recognition for VUI), Terminal

AIM:

The aim is to develop and compare Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI) for the same task, and assess user satisfaction using Python (with Tkinter for GUI and Speech Recognition for VUI) and Terminal.

PROCEDURE:

i) CLI (Command Line Interface)

CLI implementation where users can add, view, and remove tasks using the terminal.

```
import datetime

tasks = []
while True:
 print("\n--- TASK MANAGER ---")
 print(f"    Total Tasks: {len(tasks)}")
 print("1. Add Task")
 print("2. View Tasks")
 print("3. Remove Task")
 print("4. Exit")
 choice = input("Enter choice: ")
 if choice == "1":
     task = input("Enter task: ")
     priority = input("Enter priority (High/Medium/Low): ")
     tasks.append({"task": task, "priority": priority, "added":
  datetime.datetime.now().strftime("%H:%M")})
     print("Task added! ✅")
 elif choice == "2":
     if not tasks:
         print("No tasks available.")
     else:
         for i, t in enumerate(tasks, 1):
             print(f"{i}. [{t['priority']}] {t['task']}  (added at {t['added']})")
 elif choice == "3":
     num = int(input("Enter task number: "))
     if 1 <= num <= len(tasks):
         print("Removed:", tasks.pop(num - 1)["task"])
     else:
         print("Invalid number")
 elif choice == "4":
     break
 else:
     print("Invalid choice")
```

Sample Output:

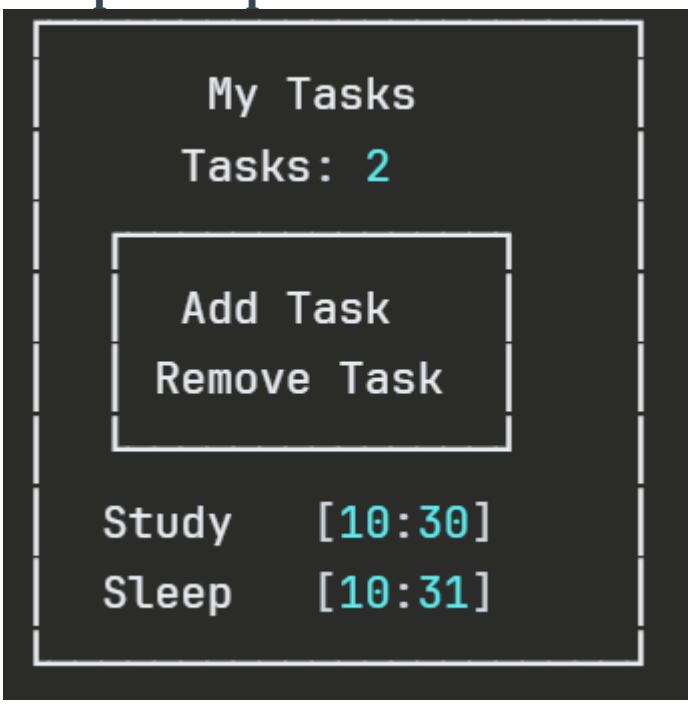
```
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter choice: 1
Enter task: Study for exam
Enter priority (High/Medium/Low): High
Task added! ✅

--- TASK MANAGER ---
    Total Tasks: 1
...
Enter choice: 2
1. [High] Study for exam  (added at 10:45)
```

ii) GUI (Graphical User Interface)

```
import tkinter as tk
from tkinter import simpledialog, messagebox, ttk
import datetime

tasks = []

def add_task():
    task = simpledialog.askstring("Add Task", "Enter your task:")
    if task:
        priority = simpledialog.askstring("Priority", "Enter priority (High / Medium / Low):")
        if priority not in ["High", "Medium", "Low"]:
            priority = "Medium"
        added = datetime.datetime.now().strftime("%H:%M")
        tasks.append({"task": task, "priority": priority, "added": added})
        update_list()
        update_count()

def remove_task():
    selected = task_list.curselection()
    if selected:
        tasks.pop(selected[0])
        update_list()
        update_count()
    else:
        messagebox.showwarning("Warning", "Select a task to remove")

def mark_done():
    selected = task_list.curselection()
    if selected:
        index = selected[0]
        tasks[index]["task"] = "✓ " + tasks[index]["task"]
        update_list()
    else:
        messagebox.showwarning("Warning", "Select a task to mark done")

def clear_all():
    confirm = messagebox.askyesno("Confirm", "Clear all tasks?")
    if confirm:
        tasks.clear()
        update_list()
        update_count()

def update_count():
    count_label.config(text=f"Total Tasks: {len(tasks)}")

def update_list():
    task_list.delete(0, tk.END)
    colors = {"High": "#ff6b6b", "Medium": "#ffa94d", "Low": "#69db7c"}
```

```
    for t in tasks:
        label = f"[{t['priority']}] {t['task']}  ⏰ {t['added']}"
        task_list.insert(tk.END, label)
        task_list.itemconfig(tk.END, fg=colors.get(t["priority"], "white"))

# Main window
app = tk.Tk()
app.title("To-Do List")
app.geometry("340x420")
app.configure(bg="#1e1e2e")

tk.Label(app, text="My Tasks", font=("Courier", 14, "bold"),
         bg="#1e1e2e", fg="#cdd6f4").pack(pady=8)

count_label = tk.Label(app, text="Total Tasks: 0", font=("Courier", 10),
                       bg="#1e1e2e", fg="#a6adc8")
count_label.pack()

btn_style = {"width": 20, "font": ("Courier", 10), "bg": "#313244",
             "fg": "#cdd6f4", "activebackground": "#45475a", "relief": "flat", "pady": 4}

tk.Button(app, text="➕ Add Task",    command=add_task,    **btn_style).pack(pady=4)
tk.Button(app, text="🗑 Remove Task", command=remove_task, **btn_style).pack(pady=4)
tk.Button(app, text="✓ Mark as Done", command=mark_done,  **btn_style).pack(pady=4)
tk.Button(app, text="✕ Clear All",    command=clear_all,   **btn_style).pack(pady=4)

task_list = tk.Listbox(app, width=38, height=10, font=("Courier", 10),
                       bg="#181825", fg="#cdd6f4", selectbackground="#45475a",
                       relief="flat", bd=8)
task_list.pack(pady=10)

app.mainloop()
```

Sample Output:



iii) VUI (Voice User Interface)

speech_recognition library for voice input and the pyttsx3 library for text-to-speech output. Make sure you have these libraries installed (pip install SpeechRecognition pyttsx3).

```
import os
import platform
import datetime

def listen():
    command = "system info"
    print("> Listening... (demo mode)")
    print(f"  You said: '{command}'")
    return command.lower()

def show_info():
    now = datetime.datetime.now().strftime("%Y-%m-%d %H:%M")
    print("\n--- SYSTEM INFO ---")
    print(f"  OS        : {platform.system()}")
    print(f"  Version   : {platform.version()[:20]}")
    print(f"  User      : {os.getlogin()}")
    print(f"  Time      : {now}")
    print("-------------------")
    print("  ✅ Voice command executed!")

if __name__ == "__main__":
    command = listen()
    if "system" in command:
        show_info()
    else:
        print("❌ Command not recognized)
```

Sample Output:

```
> Listening... (demo mode)
  You said: 'system info'

--- SYSTEM INFO ---
  OS         : Windows
  Version    : 10.0.26100
  User       : Pooja
  Time       : 2025-06-10 10:45
--------------------
```